

Building a Golf AI Swing Coach

From GolfDB to a shipped product — the label problem, the pose stack, the scoring rubric, the feedback engine, the unit economics, and a validation gate that fires before the money does.

PREPARED FOR

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DATE

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HORIZON

Phase 0 to Phase 3 · ~15 months

MVP TARGET

4–6 months

Executive Summary & How to Read This

You have a dataset, a strong technical background, and a market that has just filled up with competitors. This plan is sequenced around a single principle: **spend the least possible money proving the riskiest assumption first**, and treat every subsequent phase as conditional on the previous one clearing.

The four things that determine whether this works

- 1. The label problem.** GolfDB gives you 1,400 swings with eight labelled events and *no quality scores*. There is no ground truth for "good swing" anywhere in your data. How you manufacture that label determines your entire model architecture. Section 02.
- 2. Monocular accuracy.** Single-camera 3D is roughly 2° under laboratory conditions and materially worse in a car park at dusk. Your product must be honest about which metrics survive that. Section 03.
- 3. The capture constraint.** A browser cannot record the framerate golf requires. This breaks the naive web-first plan and forces an explicit decision. Section 06.
- 4. Retention.** Fitness apps retain roughly 3% of installs at day 30. Your scoring model can be excellent and the business can still fail here. Section 05 and Section 08.

THE RECOMMENDED PATH IN ONE PARAGRAPH

Build a **2D-metrics product first** — swing event segmentation from SwingNet, pose from RTMPose, a transparent biomechanics rubric producing per-phase sub-scores, fault flags mapped to the TPI framework, and an LLM that verbalises measured findings without ever inventing them. Ship it as **upload-only on web** to avoid the browser capture problem entirely. Validate with real golfers and paid coaches before writing a line of mobile code. Add native high-framerate capture and club tracking in Phase 2. Defer 3D until you have both a freedom-to-operate opinion on Sportsbox's patents and evidence that customers will pay for it.

What this plan deliberately does not do

- ▶ It does not attempt ground reaction forces, weight distribution or pressure — those require a force plate and cannot be inferred from video.
- ▶ It does not attempt precise wrist kinematics — that is what HackMotion's 800Hz sensor exists for.
- ▶ It does not attempt launch-monitor-grade clubhead speed or ball data from a phone camera.
- ▶ It does not build a 3D avatar in Phase 1, for patent and accuracy reasons both.

Every one of those omissions is a claim your competitors make and cannot support. Declining to make them is part of the trust positioning identified in the competitor report.

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The Label Problem

This is the hardest intellectual problem in the product and the one most competitors have quietly fudged. Solve it well and you have a defensible technical position; solve it lazily and you have another app that outputs a confident number meaning nothing.

What GolfDB actually gives you

GolfDB — McNally et al., CVPR Workshops 2019 — contains approximately **1,400 real-time swing videos** annotated with **eight swing events**: address, toe-up, mid-backswing, top, mid-downswing, impact, mid-follow-through and finish. It also carries view type (face-on or down-the-line), club, sex and player name.

The baseline model, **SwingNet**, is a MobileNetV2 backbone with a bidirectional LSTM — 5.38M parameters, roughly 10.9 GFLOPs — achieving **76.1% PCE** on event detection, rising to 91.8% when scored on six of the eight events. The public repository weights reach about 71.5% without augmentation.

THE GAP, STATED PLAINLY

SwingNet solves **temporal segmentation** — it tells you *when* the top of the backswing occurred. It says nothing whatsoever about whether that backswing was any good. GolfDB contains no quality scores, no handicaps, no outcomes, no coach assessments. Every "swing score" you have seen in a competitor app was manufactured from something, and in most cases that something is a heuristic nobody has validated.

Six ways to manufacture a quality label

APPROACH	HOW IT WORKS	STRENGTHS	WEAKNESSES
1. Pro-versus-amateur proxy	GolfDB is largely professional broadcast footage. Label pro as high quality, your own amateur captures as low.	Free, immediate, requires no annotation.	Teaches the model "looks like a TV professional." Conflates camera quality, clothing, course setting and body type with swing quality. Useless for scoring a specific fault.
2. Biomechanics-derived	Compute validated kinematic metrics from pose and club, score each against published normative ranges.	Interpretable, defensible, directly drives coaching feedback, needs no labels at all.	Depends entirely on pose accuracy. Some important metrics are unmeasurable from one camera. Normative ranges vary by body type and skill level.
3. Expert rubric annotation	Pay PGA or TPI-certified coaches to score swings against a fixed rubric.	Gold standard. Enormous credibility value in marketing and with investors.	Costs real money. Inter-rater variance on absolute scores is notoriously poor — coaches disagree, and legitimately so.
4. Outcome-linked	Pair each swing with ball flight, handicap, or launch monitor data.	Objective, and tied to what golfers actually care about. Nobody else has this at scale.	Does not exist in GolfDB — you must collect it. Noisy: good swings produce bad shots and vice versa.
5. Contrastive / self-supervised similarity	Embed swings in a learned space, score by distance from the professional manifold.	Scales without labels. Captures holistic qualities a rubric misses.	"Different" is not "worse" — Furyk, Trevino and Na are all Tour winners with unorthodox swings. Produces a number you cannot explain, which is fatal for coaching.
6. Pairwise ranking	Coaches judge pairs — "A is better than B" — and you fit a Bradley-Terry or RankNet model.	Humans are dramatically more reliable at relative than absolute judgement. Labels are cheap and fast.	Yields ranking rather than an absolute score; requires a calibration step to map onto 0–100.

RECOMMENDATION — A STAGED HYBRID, IN THIS ORDER

- 1. Backbone: approach 2.** The biomechanics rubric is your core. It is interpretable, it needs no labels to get started, and critically *it generates the coaching feedback for free* — every point deducted is a thing to work on. This is the difference between a score and a lesson.
- 2. Calibration: approach 6.** Run pairwise coach rankings over 300–500 swings and use them to fit the weights of your rubric. This is far cheaper and more reliable than asking coaches for absolute scores, and it converts subjective expertise into objective parameters.
- 3. Secondary signal: approach 5.** Use pro-similarity as a supplementary "style" indicator and for pretraining representations — never as the headline number, because you cannot explain it.
- 4. Long game: approach 4.** Instrument outcome capture from day one even though you cannot use it yet. In eighteen months this is the only moat you will have that a competitor cannot replicate by reading the same papers.

This directly answers the question you posed at the outset. **Neither pure similarity nor pure rubric is right.** The rubric is the trustworthy, explainable core; similarity is complementary and secondary; coach ranking is the calibration anchor that makes the rubric defensible rather than arbitrary.

Datasets to acquire beyond GolfDB

DATASET	SOURCE	WHAT IT GIVES YOU
GolfDB	github.com/wmcnally/golfdb	Your base. 1,400 swings, 8 events, view labels. Important: the Kaggle 160×160 mirror is adequate for training SwingNet but far too low-resolution for quality pose estimation — source higher-resolution clips for the pose work.
GolfPose	ICPR 2024, Lee et al.	The key academic reference for club-aware pose. Its GolfSwing set annotates body keypoints <i>plus five club keypoints</i> , reaching club-keypoint mAP of 0.918–0.956. Its post-processing step uses OpenCV line-segment detection to correct the hosel keypoint, cutting pixel-error standard deviation from 33.89 to 22.39.
GolfMate	MDPI Applied Sciences 2023	Pose-refinement network plus an explainable swing embedding for self-training, built from a professional database of 16 PGA Tour players. A direct template for the similarity component of your hybrid.
Wrist-IMU golf study	arXiv 2506.17505 (2025)	Reconstructs full-body 3D swings using WHAM on monocular video across 1,046 swings and 139,503 frames from down-the-line, face-on and rear views. Confirms WHAM is viable for golf specifically — the single most useful validation for your 3D path.
Roboflow Universe club sets	Roboflow	Ready-made YOLO-format club and clubhead detection data — "golf-club-tracking" at roughly 6,750 images, "Golf Club Detection" at roughly 8,577. Directly in your existing skillset and the fastest route to a club detector.

THE GAP THAT IS YOUR OPPORTUNITY

No public dataset pairs golf swings with handicap and launch-monitor outcomes at any scale. GolfTec has 14 million swings behind a studio wall. That absence is precisely where a proprietary data asset can be built — see Section 07.

Pose Estimation & the Accuracy Ceiling

2D pose — start here, and possibly stay here

MODEL	LICENCE	PROFILE	VERDICT FOR THIS PRODUCT
MediaPipe / BlazePose	Apache 2.0	33 kpts, fastest, on-device	Use for on-device pre-checks — framing, lighting, full body in frame, correct angle. Lowest accuracy of the group. Note that a 33-keypoint output is the signature that gives away most competitor apps.
YOLOv8 / YOLO11-pose	AGPL-3.0	Real-time, edge-friendly	Your existing skillset and easy Jetson deployment. Check the licence carefully — Ultralytics AGPL has real commercial implications; a commercial licence is available.
RTMPose / RTMW	Apache 2.0	RTMPose-x ≈75.8% AP on COCO	Recommended for the cloud pipeline. Best accuracy-to-speed ratio available, permissively licensed — and Swing Catalyst, a premium commercial golf system, uses exactly RTMDet-M plus RTMPose-X with SimCC. That is strong independent validation.
ViTPose / ViTPose++	Apache 2.0	Transformer, high accuracy	A premium-accuracy backend option. Heavier and more expensive per inference.
Sapiens (Meta)	Research/CC	0.3B–2B params; Sapiens-0.3B beats ViTPose++-L by ~7.9 AP	Highest accuracy available. Reserve for difficult or occluded frames, or as a premium tier. Check licence terms for commercial use.

Monocular 3D — and an honest assessment of whether it works

If you pursue 3D, the relevant models are **WHAM** and **TRAM** for world-grounded motion from monocular video, alongside **4D-Humans** / **HMR2.0**, **SMPLer-X**, **NLF** and **PHALP** for per-frame or tracked SMPL mesh recovery. WHAM runs at roughly 200fps, is state of the art on 3DPW and EMDB, and fuses 2D keypoints with video features, camera motion and foot contact. It is already validated on golf by the 2025 wrist-IMU paper.

THE ACCURACY REALITY CHECK — READ THIS BEFORE PROMISING ANYONE 3D

Sportsbox's own study reports roughly **2° absolute error against AMM3D** on turn, bend and side-bend at key positions — but explicitly only under controlled conditions.

Independent markerless-versus-marker studies in adjacent movements set the honest ceiling. MMPose-based countermovement jump analysis achieved correlations above 0.98 on temporal variables, which is excellent. But Theia3D markerless tracking in dynamic boxing showed **errors above 3cm at the elbow and under 2.5cm at shoulder and wrist** — centimetre-scale joint-centre errors that propagate into several degrees of angular error, and get worse for fast-moving and occluded segments. A golf downswing is both.

Conclusion. Single-camera 3D is adequate for directional coaching feedback on large angles — pelvis and thorax turn, sway, side bend. It is *not* adequate for fine wrist kinematics or anything requiring sub-degree precision. If you build the 3D tier, strongly consider **two-camera capture** — face-on plus down-the-line, which is precisely what GolfTec's OptiMotion does and what Uplift Labs requires.

Club tracking — hard, and therefore a differentiator

The clubhead travels at over 100mph. It is thin, specular, heavily motion-blurred, and frequently occluded by the golfer's body. Most competitor apps simply do not attempt it, which is why club tracking is worth solving.

- ▶ **Primary approach** — a dedicated YOLO club and clubhead detector trained on the Roboflow datasets, combined with a temporal trajectory model to interpolate through blurred and occluded frames.
- ▶ **Refinement** — GolfPose's five-club-keypoint head, plus its line-segment-detection correction for the hosel keypoint.
- ▶ **Classical fallback** — frame differencing plus Hough or LSD line detection for the shaft, which is more robust to blur than a learned detector on a thin object.

Sequencing: start body-only. Body pose is robust and delivers most of the coaching value. Add club tracking as a fast-follow in Phase 2. Do not promise launch-monitor-grade clubhead speed or path from a phone camera under any circumstances.

Capture standardisation — the cheapest accuracy win available

Far more product failures come from bad input than from bad models. Three non-negotiables:

1. **Support exactly two angles** — face-on and down-the-line, matching GolfDB's own view labels. Provide on-screen alignment guides. Reject or flag off-axis uploads rather than analysing them badly.
2. **Enforce a framerate floor.** Below 120fps the impact region is unusable. Detect framerate on upload and warn explicitly.
3. **Handle scale ambiguity.** Monocular video has no absolute scale. Recover it from a known reference — club length, the golfer's entered height, or standard stance width. **Report angles as primary** since they are scale-invariant, and give linear measurements in inches only when calibration succeeded.

The Biomechanics Rubric

This section defines what your score is actually made of. It should be published openly — transparency about your rubric is a trust asset, and it is not the part a competitor can easily copy anyway, because the value is in the calibration, not the list.

What the literature supports

VARIABLE	FROM ONE CAMERA?	EVIDENCE AND COACHING VALUE
X-factor & X-factor stretch hip-shoulder separation	Yes	The most-cited driver of clubhead speed. Nesbit et al. found peak X-factor and X-factor at impact correlate strongly with clubhead speed, with median r around 0.90–0.94 in elite swings, peaking in early downswing. Caveat: large-sample correlations are partly inflated by skill heterogeneity.
Kinematic sequence pelvis → thorax → arm → club	Yes	The signature of efficient energy transfer, measured as the ordering of peak angular velocities. High explanatory value, highly teachable, and exactly what K-Vest sells for hundreds of dollars. Your strongest sub-score.
Pelvis & thorax rotation ranges	Yes	Well-established normative ranges; straightforward from pose.
Spine angle & side bend	Yes	Directly maps to several TPI faults including reverse spine angle and S/C posture.
Early extension	Yes	Detectable geometrically via the tush line. Appears in roughly 70% of amateurs and almost no Tour players — an unusually high-signal, highly teachable fault. Prioritise this one.
Sway, slide, head movement	Yes	Simple translational measures, robust to pose noise, easy to explain to a golfer.
Tempo ratio ~3:1 backswing to downswing	Yes — trivially	Derived directly from SwingNet event timings with no pose required at all. Cheap, reliable, popular and widely understood. Ship this in your very first version. Note that Swing Lab already ships exactly this, including the 2:1 chip ratio.
Swing plane	With club tracking	Requires club detection; defer to Phase 2.
Lag / casting	Approximately	Partially inferable with good club tracking. Treat as directional, not precise.
Lead wrist flexion/extension	No	Small, fast, self-occluding, rotating about its own axis. This is HackMotion's entire product category. Do not claim it.
Ground reaction forces, weight transfer	No	Requires a force plate or pressure mat. Cannot be inferred from video. Any competitor claiming "weight transfer" from a phone camera is inferring it from centre-of-mass proxy at best.

The TPI framework as your fault taxonomy

Adopt the Titleist Performance Institute's twelve swing characteristics as your fault vocabulary. Two reasons: they are geometric and therefore detectable from pose plus reference lines, and — more importantly — every certified coach already knows them, which makes your output instantly legible to the professionals you want to recruit and sell to.

THE TPI BIG 12 — YOUR FAULT-DETECTION TARGETS

S-posture · C-posture · Loss of posture · Flat shoulder plane · Early extension · Over-the-top · Sway · Slide · Reverse spine angle · Hanging back · Casting and scooping · Chicken winging

Recommended scoring architecture

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Segment	Measure	Sub-score	Flag faults	Aggregate
SwingNet finds the 8 events, dividing the swing into phases	RTMPose extracts joint angles and velocities per phase	Each phase scores 0–100 from its weighted metric set	TPI Big 12 detection with per-fault severity	Transparent overall score with full attribution

Three design rules that matter more than the specific weights:

- ▶ **Always show what drove the score down.** A number without attribution is worthless for coaching and destroys trust the first time a golfer disagrees with it. Explainability *is* the product.
- ▶ **Anchor "good" ranges to published Tour norms, then adjust with your coach-ranking calibration.** Raw Tour ranges are the wrong target for a 20-handicapper and will produce demoralising scores.
- ▶ **Report confidence alongside every metric.** When pose confidence is low or the camera angle is off, say so rather than producing a confident wrong number. No competitor does this and it is a meaningful trust differentiator.

The Feedback Engine

This plays directly to your RAG and agentic LLM background — but the architectural discipline matters more than the model choice. The central rule: **the LLM never measures anything.**

Architecture

DETERMINISTIC	RETRIEVAL	GENERATIVE	VALIDATION
Rules engine	RAG over drills	LLM verbalises	Consistency check
Metrics + thresholds → structured findings: fault, phase, severity, measured vs target	Vetted coaching corpus indexed by fault; retrieve for top 1–2 detected faults	Explains findings and selects drills — never invents a number	Reject any output contradicting the measured metrics

The LLM receives structured JSON findings, not video and not raw keypoints. It cannot hallucinate a measurement it was never given. This eliminates the primary failure mode in the entire category — and it is the reason to resist the tempting shortcut of feeding frames directly to a vision-language model. Current VLMs are unreliable on fine golf biomechanics and will confidently invent angles. Use them, if at all, for camera-setup quality assurance, never as the source of a metric.

The drill corpus is a real asset

Index vetted drills keyed by fault, sourced from coaches you pay or partner with. This doubles as a content moat and as the hook for a coach marketplace later. Performance Golf's model — bundling AI analysis with a large instructional library and a coaching upsell — shows the commercial logic.

Retention is designed here, not bolted on later

THE LOOP THAT KEEPS PEOPLE PAYING

- **One focus at a time.** Prescribe a single fault, never a list of nine. SWEE markets explicitly on breaking things into manageable pieces, and the best review of Swing Coach praises its *restraint* in stopping golfers over-working a swing thought. This is a coaching insight, not a UX preference.
- **Trend deltas as the hero metric.** "Your early extension improved 40% over three weeks" is worth more than any single score. It is also the thing a golfer will pay to keep seeing.
- **Prompt the second swing.** The single most important conversion event is a user uploading a *second* swing. Design the entire first-run experience around producing it.
- **Seasonal re-engagement.** Golf collapses in winter across your core markets. Build indoor and simulator-friendly flows, and plan a deliberate spring reactivation campaign.

Architecture & Unit Economics

The web capture problem — an important correction to your plan

You proposed finalising the product on web, then building iOS and Android. That instinct is right for most of the system and wrong for one critical component, and the distinction determines your v1 scope.

A BROWSER CANNOT CAPTURE THE FRAMERATE GOLF REQUIRES

- `getUserMedia` and `MediaRecorder` treat `frameRate` as best-effort and silently cap it — MDN notes the browser may reduce framerate if constrained.
- iOS Safari caps camera capture at roughly 30fps and 720p and exposes no API to the iPhone's 120 or 240fps slow-motion mode — see WebKit Bugzilla #179994 and Apple Developer Forums.
- Android's 120 and 240fps modes require the native Camera2 high-speed API; standard MediaRecorder is stuck at 30fps.
- Even native code under-delivers — AVFoundation at 120fps often yields around 70fps in practice.

At 30fps a clubhead moving over 100mph traverses several feet between frames. The impact region is simply not recoverable. **Slow-motion capture is reachable only through native code.**

THE RESOLUTION — SMALLEST VIABLE V1

Build everything except capture on web. The analysis engine, results interface, coach dashboard, account management and billing are all genuinely better built once in React and FastAPI.

For v1, make capture upload-only. The golfer films with their phone's *native* slow-motion camera and uploads the file. This is exactly the workaround Sportsbox's Android users already tolerate. It validates the entire analysis, scoring and feedback loop with zero mobile development. Then build a **React Native** capture app in Phase 2 — it shares your React components and gives one codebase, with native modules for Camera2 and AVFoundation high-speed capture. Flutter would fork your skillset; dual native is twice the work. Go fully native only if React Native's high-speed camera modules prove inadequate in testing.

Reference pipeline

CAPTURE	TRANSPORT	INFERENCE	FEEDBACK	DELIVERY
Native slow-mo On-device MediaPipe pre-checks: framing, lighting, angle, framerate	Presigned upload Direct to S3 or GCS, then queue via SQS or Redis	GPU worker SwingNet → RTMPose → optional WHAM → club → metrics → rubric	Rules → RAG → LLM Structured findings verbalised and validated	Push to client Results stored, warehoused to Snowflake for the flywheel

FastAPI as the orchestration layer plays to your existing strengths, with Databricks or Snowflake as the analytics warehouse feeding the data flywheel described in Section 07.

Real infrastructure costs

LINE ITEM	COST	NOTES
Serverless GPU	\$0.58–3.95/hr	RunPod serverless: A100 80GB roughly \$1.07–3.49/hr, L40S around \$0.91/hr, RTX 4090 around \$0.58/hr. Modal H100 around \$3.95/hr effective, billed per second with sub-second cold starts via container caching. Replicate A100 around \$8.28/hr but zero operational overhead.
Per swing analysis	\$0.005–0.03	Assuming 10–30 seconds on an L4 or L40S class card. Trivial against a \$9.99 subscription — provided you scale to zero . An idle A100 left running costs roughly \$800/month.
Video storage	~\$0.023/GB-mo	A 5–10 second 1080p 120fps clip runs 30–80MB. Transcode and downsample after analysis; lifecycle raw uploads to cold storage or delete them.
Egress	Variable	The sneaky cost. Keep processing in-region with storage.
LLM feedback	~\$0.01/swing	Structured findings in, short coaching text out — a small prompt and a small completion.

UNIT ECONOMICS AT A \$9.99 SUBSCRIPTION

Roughly **\$0.02 GPU + \$0.01 storage and egress + \$0.01 LLM** per analysis. At 30 analyses per user per month that is approximately **\$1.20 per user per month in COGS against \$9.99 in revenue** — an 88% gross margin. The economics are healthy. **Churn, not COGS, is what determines whether this business works.**

Recommendation: Modal for the best developer experience with per-second billing and scale-to-zero, or RunPod serverless for lowest raw cost. Run FastAPI and the queue on cheap CPU instances. Cache model weights on platform storage — cold-start weight downloads can add fifteen seconds to a request. Target under 10–20 seconds end-to-end with an async "processing" state; true real-time is unnecessary and expensive.

Cut cost with on-device pre-checks

Run MediaPipe or a small Core ML / TF Lite pose model on-device purely to validate the capture *before* upload — full body in frame, correct angle, adequate lighting, sufficient framerate. This avoids paying for cloud inference on unusable video and dramatically improves perceived quality, since the user is corrected in the moment rather than told twenty seconds later that their swing could not be analysed.

Data Moat, Operations & IP

The flywheel

Every consented swing plus any outcome the golfer supplies — self-reported handicap, later launch-monitor pairing, eventually ball flight — feeds a dataset that does not exist publicly and that no competitor can obtain by reading the same papers you did. GolfTec's 14 million swings show what this looks like at maturity. Instrument the capture from day one in Snowflake or Databricks, even in Phase 0 when you cannot yet use the data.

Recruiting coaches cheaply

You need three to five PGA or TPI-certified coaches for two jobs: building the pairwise ranking calibration set from Section 02, and authoring drills for the RAG corpus. Offer equity, revenue share, or per-swing micro-payments rather than consulting rates. Their names then become marketing credibility — this is precisely the Sportsbox advisory playbook, which recruited Foley, Leadbetter and Larkin early and used them as both product input and proof of seriousness.

Operating from Pakistan

GENUINE ARBITRAGE, WITH TWO FRICTIONS TO SOLVE

Data collection. Islamabad and Rawalpindi have established facilities — Islamabad Golf Club, Rawalpindi Golf Club and associated courses, plus driving ranges. You can film consented amateurs across a full skill range at 120 or 240fps for a fraction of US labour cost. Building the dataset and pipeline locally while selling into the US and UK is a real structural advantage.

Friction 1 — payments. Stripe does not cleanly support receiving global card payments into Pakistan. Solve this by incorporating a **US LLC via Stripe Atlas or a UK Ltd**, or by using a merchant-of-record such as **Paddle or Lemon Squeezy** which handles global tax and VAT and pays out to Pakistan. App Store and Play Store payouts also require the foreign entity's banking.

Friction 2 — support hours. US customer support from PKT is a real operational constraint. Plan asynchronous support with clear response-time expectations rather than pretending to offer live coverage.

Patent exposure

RESOLVE BEFORE BUILDING ANY 3D FEATURE

Sportsbox holds granted US patents on markerless 3D golf swing biomechanical analysis. **US 11,640,725** claims generating 3D avatars from images together with a user-selectable watchlist of kinematic parameters, club tracking and pre-shot-routine machine learning. GolfTec and the launch-monitor manufacturers hold further motion-measurement and club-tracking IP.

Two de-risking moves. First, lead with a **2D metrics plus rubric score plus LLM feedback** product, which sits well outside the 3D-avatar-plus-watchlist claims. Second, build on openly published methods — RTMPose, WHAM, GolfPose — and open datasets, which strengthens your position considerably. Commission a freedom-to-operate opinion from a patent attorney before Phase 3, not after.

Monetisation & Go-to-Market

Pricing

Freemium with a hard value gate. Free tier limited to a small number of scored swings per month — Sportsbox uses five, which is enough to prove value and not enough to live on. **Pro at \$9.99/month or \$59.99–99.99/year.** You will not command a premium; the category is anchored and several competitors are free or one-time purchases.

PUSH ANNUAL, HARD

Annual subscribers retain **44.1% at twelve months against 17.5% for monthly**. That 2.5–3× LTV difference dwarfs any realistic improvement in conversion rate or model accuracy. Price the annual plan aggressively — two to three months free — and default to it in the paywall. Also consider a one-time detailed swing report for non-subscribers; SwingDraw reaching #1 paid app in Sports at \$4.99 with no subscription proves a meaningful segment actively rejects recurring billing.

Sequencing B2C and B2B

Start B2C for speed of validation and because you own the feedback loop. But design the coach and academy tier from the outset — it carries higher ARPU, lower churn, and each coach brings students. Note from the competitor report that **swingiq.app now occupies the coach-facing lane**, so it is no longer greenfield.

The most defensible commercial path is the third one: **white-label the coaching brain to launch-monitor and simulator companies**. Foresight already licences body data from Sportsbox; Full Swing licences video from Onform. These companies have distribution and hardware but no biomechanics layer, and they have demonstrated they will buy rather than build.

Acquisition channels

CHANNEL	APPROACH AND NOTES
Reddit — r/golf, r/golfeq	High-intent, sceptical, and brutal on overclaiming. An honest accuracy post would perform unusually well here precisely because the category is full of hype. This is your best Phase 0 test channel.
Golf YouTube	MyGolfSpy, Rick Shiels, Me and My Golf — note the last already runs Swing Coach, so treat them as a competitor rather than a partner. Data-driven reviewers are the natural fit for an accuracy-led product.
SEO / content	Non-optional. DeepSwing runs a "best golf swing analyzer apps in 2026" article that ranks itself first and updates constantly. You will have to fight this battle.
GolfWRX forums	Equipment-obsessed, technically literate, highly influential on purchase decisions.
Driving range partnerships	QR codes at the tee line. Cheap, local, and a natural pairing with your Pakistan-based data collection — the same relationships serve both purposes.
App Store economics	Enrol in the Small Business Program for 15% rather than 30% while under \$1M annual revenue. Initiate web subscriptions through Stripe or Paddle to avoid the store cut entirely.

Risks & the Phase 0 Validation Gate

The honest case against building this

- ▶ **The category filled up while you were researching it.** A dozen apps now do the headline concept. Differentiation must come from accuracy, data and retention — all of which are harder and slower than shipping a score.
- ▶ **In-the-wild accuracy destroys trust fast.** One obviously wrong diagnosis and the golfer stops believing anything the app says. Bad angles, poor lighting and low framerate are the norm, not the exception.
- ▶ **Detecting a fault is not fixing it.** The gap between a correct fault list and an actual lower handicap is enormous, and it is where most instruction technology quietly fails.
- ▶ **The ceiling case is sobering.** SwingVision — better funded, more objective output, five years in — sits at \$2.7M revenue and roughly \$3M annual losses. Plan for a solid business, not a rocket ship.
- ▶ **Seasonality.** Northern-hemisphere golf collapses in winter. Mitigate through indoor and simulator positioning and the off-course cohort.
- ▶ **Liability.** Avoid medical and injury claims entirely — TPI screening touches injury risk and you should not follow it there. Position as performance and entertainment, with clear terms of service.

Phase 0 — the cheap experiment that comes before everything

DO THIS BEFORE WRITING PRODUCTION CODE

Build the upload-only web MVP with 2D metrics, tempo ratio, three to five TPI faults and grounded LLM feedback, running on GolfDB plus 100–300 swings you collect locally. Put it in front of r/golf or a golf YouTuber behind a \$9.99 Stripe wall.

Measure three things: do people upload a *second* swing; will they pay; and do paid coaches rate the feedback as correct.

KILL AND PIVOT CRITERIA — DECIDE THESE NOW, NOT LATER

- If **fewer than 15–20% of activated users return for a second swing**, the retention loop is broken. Fix it before building anything else — no amount of model accuracy compensates.
- If **coaches flag the feedback as wrong more than 20% of the time**, accuracy is not yet good enough to charge for. Return to the rubric and pose pipeline.
- If **nobody pays**, test the coach-pays and one-time-purchase models before concluding the market is closed — both are proven inside the competitive set.

Cost: roughly \$1,000–3,000 and a few weeks. This de-risks the entire build and it is the highest-return work in this document.

Phased Roadmap

PHASE 0 • WEEKS 1-6 • ~\$1-3K

Validation

Fork GolfDB and SwingNet. Add MediaPipe or RTMPose 2D pose. Compute tempo ratio plus three to five TPI faults and a provisional rubric score. Wire the rules → RAG → LLM feedback layer on FastAPI. Build an upload-only React page. Collect 100-300 local swings at Islamabad Golf Club or a driving range. Ship to r/golf and one YouTuber behind a \$9.99 Stripe wall.

Gate: second-swing return $\geq 15-20\%$; coach-audited feedback error $< 20\%$.

PHASE 1 • MONTHS 2-5

Web MVP

Harden the RTMPose cloud pipeline on Modal or RunPod serverless with scale-to-zero. Build per-phase sub-scores and the explainable score interface with confidence reporting. Add accounts and billing via Paddle or Stripe through a US or UK entity. Instrument event capture into Snowflake or Databricks. Recruit three to five coaches for pairwise ranking calibration and drill authoring.

Gate: paying users retained past month three; rubric calibrated against coach rankings.

PHASE 2 • MONTHS 6-9

Native capture & club tracking

React Native capture app with native high-speed camera modules for AVFoundation and Camera2, plus on-device MediaPipe pre-checks. Fine-tune a YOLO-pose model with a GolfPose-style club-keypoint head on Roboflow and GolfPose data. Add swing plane and kinematic sequence sub-scores.

Gate: mobile capture quality measurably better than user-uploaded video.

PHASE 3 • MONTHS 9-15 • GATED ON WILLINGNESS TO PAY

3D tier & B2B

WHAM or TRAM single-camera 3D — or two-camera face-on plus down-the-line for accuracy — **only after a freedom-to-operate review against Sportsbox's patents**. Launch the coach and academy tier. Pursue launch-monitor and simulator white-label integrations. Begin outcome-linked labelling through handicap and launch-monitor pairing to build the data moat.

Gate: FTO opinion in hand; demonstrated demand for 3D at a price that covers its cost.

Immediate next actions

1. **Run a proper name and trademark search.** Every obvious "Swing" name is taken. Budget for this before building any brand assets.
2. **Download GolfDB at full resolution**, not the Kaggle 160×160 mirror, and reproduce the SwingNet baseline to confirm your pipeline.
3. **Book range time locally** and film 100 swings across skill levels at 240fps. This is your Phase 0 dataset and it costs almost nothing.
4. **Approach two coaches** with an equity or micro-payment offer for calibration labelling.

5. Read US 11,640,725 in full before making any 3D architecture decisions.

THE ONE-SENTENCE VERSION

Build the 2D rubric product upload-only on web, prove that golfers come back for a second swing and that coaches agree with your diagnoses, and only then spend money on native capture, club tracking and 3D — because in a category this crowded, **provable accuracy and a retention loop that works are the only durable advantages available.**

Key sources

Datasets & models — GolfDB: github.com/wmcnally/golfdb, arXiv 1903.06528 · GolfPose: ICPR 2024, minghanlee.github.io · GolfMate: MDPI Applied Sciences 2023 · Wrist-IMU golf: arXiv 2506.17505 · RTMPose: arXiv 2303.07399 and MMPose · WHAM: arXiv 2312.07531 · TRAM: arXiv 2403.17346 · Sapiens: arXiv 2408.12569 · Roboflow Universe golf club datasets

Accuracy & biomechanics — help.sportsbox.ai/sportsbox-ai-accuracy · Nesbit et al. on X-factor and clubhead speed · TPI Big 12 swing characteristics · Swing Catalyst markerless pipeline documentation

Platform constraints — MDN `MediaTrackConstraints.frameRate` · WebKit Bugzilla #179994 · Apple Developer Forums on AVFoundation high-speed capture

Economics — RunPod, Modal and Replicate published GPU pricing · RevenueCat State of Subscription Apps · Business of Apps health & fitness retention benchmarks · US Patent 11,640,725